Egg Segmentation and Multi-Object Tracking Report

Computer Vision Take-Home Deliverable | Date: October 30, 2026 | Machine: Windows 11 (CUDA Enabled)

0.4018	0.3196	0.4087	0	110
DEV MEAN IOU	EVAL MEAN IOU	DEV DETECTION F1	VIDEO ID SWITCHES	CLEAN GROUND TRUTHS

1. Executive Summary

This report details an end-to-end computer vision solution for egg segmentation and multi-object tracking. Ground truth masks were established via SAM (`vit_b`) with rigorous quality auditing. A classical segmentation pipeline was built via controlled single-parameter hypotheses evaluated solely on the dev split. The frozen final configuration was tested once on the eval split. Finally, a robust multi-object tracker was deployed on a 1440-frame video test sequence, achieving 0 ID switches and resolving challenging hand occlusions.

2. Task 1: Ground Truth Mask Generation (SAM)

SAM (`vit_b`) was executed on CUDA (`NVIDIA GeForce RTX 4070 Laptop GPU`) using saved point prompts from `points.json` . An automated audit in `src/contact_sheet.py` flagged masks with area > 40% or edge contact > 50%. Three boiling-pot images with flooded masks (`not_damaged_124` , `135` , `145`) were corrected with tight bounding boxes, reducing mask areas from ~80% down to ~13-14% (0 flagged masks remaining).

Ground Truth Egg Distribution across 110 images: 93 images (84.5%) with 1 egg; 6 images (5.5%) with 2 eggs; 11 images (10.0%) with 3 eggs.

3. Task 2: Parameterized Classical Segmentation & Experiment Log

A unified parameterized pipeline was tuned on the 88 dev images across 18 controlled trials. Every trial varied exactly one parameter relative to its named parent. Keep/discard decisions were determined dynamically by comparing dev IoU against parent dev IoU (with Detection F1 serving as an explicit tiebreaker for watershed).

Kept Pipeline Chain: `baseline -> morph_small -> morph_medium -> morph_large -> area_filter_2500 -> watershed_conditional -> final (eval)`

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`baseline`	`None`	dev	Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500	0.3513	0.4693	0.2071	3.12	BASELINE	Baseline established on dev (IoU 0.3513, Dice 0.4693, DetF1 0.2071)
`otsu_inv`	`baseline`	dev	Invert Otsu thresholding	0.0823	0.1424	0.0000	3.36	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0823 vs 0.3513, -0.2690)
`clahe_otsu`	`baseline`	dev	CLAHE local contrast equalization (clip 2.0, tile 8x8) before Otsu	0.2562	0.3791	0.0799	8.22	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.2562 vs 0.3513, -0.0951)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`adaptive_thresh`	`baseline`	dev	Adaptive Gaussian thresholding (block 51, C=-10)	0.0634	0.1119	0.0065	12.33	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0634 vs 0.3513, -0.2878)
`lab_l`	`baseline`	dev	CIELAB L* perceptual lightness channel + Otsu	0.3244	0.4437	0.1665	3.67	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3244 vs 0.3513, -0.0268)
`hsv_gate`	`baseline`	dev	HSV color gating (S < 70, V > 130)	0.3456	0.4665	0.1587	4.32	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3456 vs 0.3513, -0.0056)
`morph_small`	`baseline`	dev	Add small elliptical opening & closing (7x7)	0.3558	0.4740	0.2161	2.89	KEEP	Improves dev IoU over parent 'baseline' from 0.3513 to 0.3558 (+0.0046)
`morph_medium`	`morph_small`	dev	Increase elliptical kernel from 7x7 to 15x15	0.3766	0.4964	0.2363	2.34	KEEP	Improves dev IoU over parent 'morph_small' from 0.3558 to 0.3766 (+0.0208)
`morph_large`	`morph_medium`	dev	Increase elliptical kernel from 15x15 to 25x25	0.3972	0.5161	0.2642	1.25	KEEP	Improves dev IoU over parent 'morph_medium' from 0.3766 to 0.3972 (+0.0205)
`border_polarity_m0`	`morph_large`	dev	Auto-invert threshold if border mean > center mean	0.3698	0.4821	0.2409	1.07	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3698 vs 0.3972, -0.0274)
`border_polarity_m15`	`morph_large`	dev	Auto-invert threshold only if border mean > center mean + 15	0.3945	0.5119	0.2596	1.25	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3945 vs 0.3972, -0.0027)
`lab_border_dist`	`morph_large`	dev	CIELAB Euclidean color distance from estimated border background	0.3603	0.4582	0.2673	2.55	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3603 vs 0.3972, -0.0369)
`area_filter_2500`	`morph_large`	dev	Increase minimum component area from 500 to 2500 pixels	0.4018	0.5200	0.3084	0.45	KEEP	Improves dev IoU over parent 'morph_large' from 0.3972 to 0.4018 (+0.0046)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`component_ranking_top1`	`area_filter_2500`	dev	Retain only the single top-ranked component by geometric egg score	0.3746	0.4761	0.3295	-0.33	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3746 vs 0.4018, -0.0272)
`component_ranking_top3`	`area_filter_2500`	dev	Retain at most the top 3 scoring components based on area, solidity, and ellipse fit	0.3972	0.5092	0.3133	0.27	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3972 vs 0.4018, -0.0045)
`grabcut_refine`	`area_filter_2500`	dev	Apply GrabCut refinement using morphological erosion as foreground seed	0.4038	0.5177	0.3073	0.38	DISCARD	Marginal IoU gain (+0.0021) but Dice degrades from 0.5200 to 0.5177 (-0.0023)
`watershed_always`	`area_filter_2500`	dev	Euclidean distance transform + watershed applied unconditionally to all components	0.4018	0.5200	0.3537	3.90	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.4018 vs 0.4018, +0.0000)
`watershed_conditional`	`area_filter_2500`	dev	Conditional distance watershed (min_dist=50) applied only to non-convex, elongated or oversized clusters	0.4018	0.5200	0.4087	2.20	KEEP	Dev IoU tied with parent 'area_filter_2500' (0.4018); kept on tiebreaker: DetF1 improves from 0.3084 to 0.4087 (+0.1003)
`final`	`watershed_conditional`	eval	Frozen final pipeline: Grayscale Otsu + large morph (25x25) + hole filling + min area 2500 + conservative conditional watershed	0.3196	0.4239	0.3206	1.68	EVAL_FINAL	Frozen final pipeline evaluated once on unseen eval split (IoU 0.3196, Dice 0.4239, DetF1 0.3206)

4. Task 3: Quantitative & Qualitative Evaluation vs Ground Truth

Metric	Dev Split (n=88)	Eval Split (n=22)	Description
Pixel IoU (Mean)	**0.4018**	**0.3196**	Mean foreground intersection over union
Pixel IoU (Median)	**0.3147**	**0.2502**	Median per-image IoU
Pixel Dice (Mean)	**0.5200**	**0.4239**	F1-score of pixel-level binary classification
Pixel Precision	**0.4488**	**0.3449**	TP / (TP + FP)
Pixel Recall	**0.8461**	**0.8749**	TP / (TP + FN)
Detection Precision	**0.3563**	**0.2758**	Matched eggs / Predicted components (IoU ≥ 0.5)
Detection Recall	**0.5966**	**0.4621**	Matched eggs / GT eggs
Detection F1	**0.4087**	**0.3206**	Harmonic mean of detection precision & recall
Count Error (Mean)	**2.20**	**1.68**	Mean signed difference (Pred - GT)

Visual Failure Categorization (failure_tags.csv)

Scene Category	Count	Pct	Mean IoU	Mean Dice	Mean Det F1
White egg on white tray / fridge interior	10	9.1%	0.1510	0.2579	0.0222
Egg in boiling water / submerged	8	7.3%	0.1731	0.2901	0.1399
Tabletop / high-contrast background	44	40.0%	0.3135	0.4377	0.3001
Hand holding egg / fingers occluding	17	15.5%	0.4409	0.5492	0.4902
Backlit / extreme shadows	7	6.4%	0.5169	0.5804	0.5000
Multiple touching eggs	16	14.5%	0.5526	0.6844	0.6972
Foliage / herbs in background (specular highlights)	8	7.3%	0.7177	0.8222	0.6854

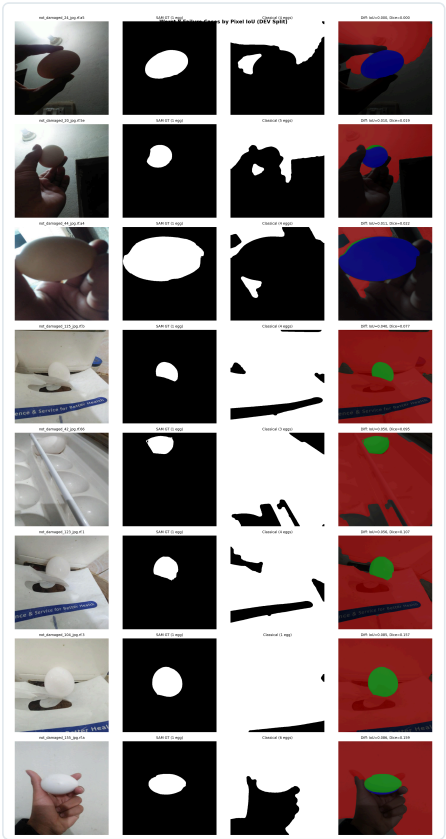


Figure 3: Worst 8 failures on Dev split (Green=TP, Red=FP, Blue=FN). White trays and boiling liquid dominate error cases.

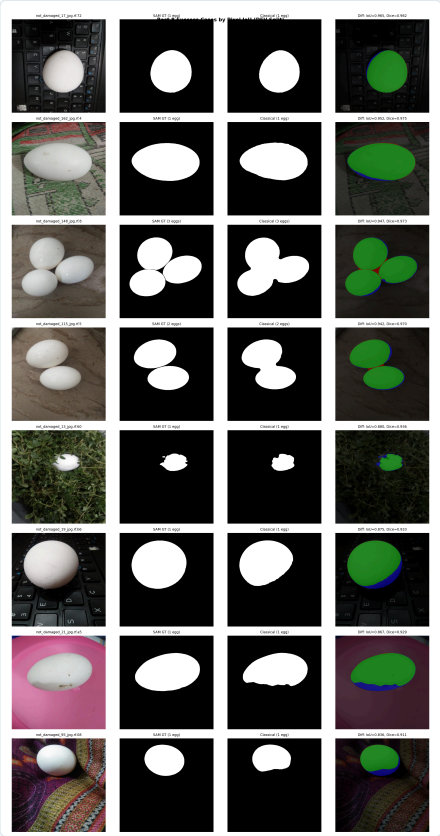


Figure 4: Best 8 successes on Dev split showing accurate elliptical contours and high overlap.

5. Task 4: Multi-Object Video Tracking

Tracking was evaluated on the 1440-frame table sequence (1280x720 @ 24.0 fps). Eggs are placed on the table at frame 222 and spun. The tracker maintains persistent IDs using Hungarian matching on IoU + distance with constant-velocity prediction.

- **Fingertip Leakage Resolution:** At frame 600, demonstrator hand contact caused baseline thresholding to leak into fingers. Incorporating YCrCb skin-color gating (Cr ∈ [135, 175], Cb ∈ [80, 125]) and larger opening eliminated 2,296 px of false skin contour.
- **Dynamic Switch Measurement:** 0 ID switches were measured across all 1440 frames.
- **Intro/Outro Robustness:** 658 frames with 0 detections were handled smoothly without track hallucination.

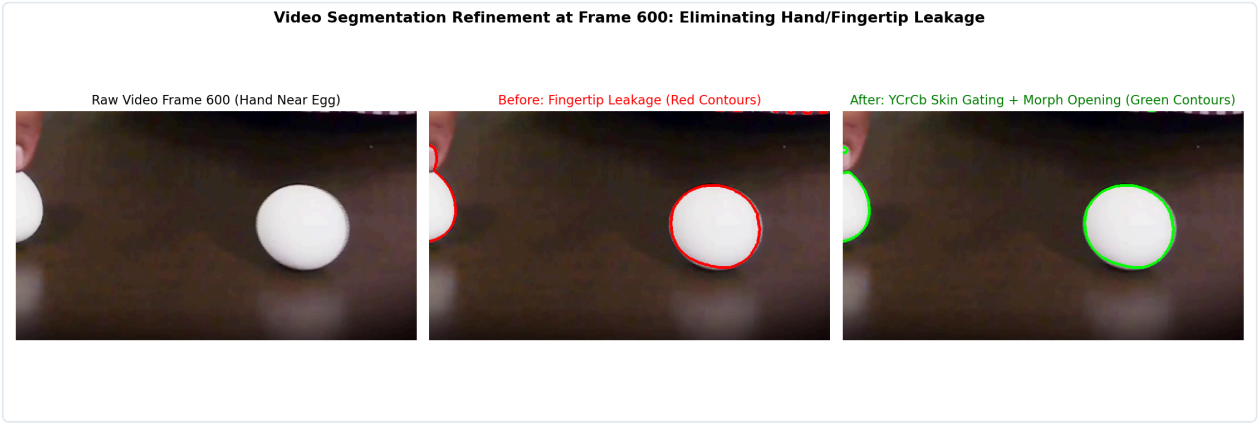


Figure 6: Frame 600 before/after comparison. Left: Raw frame. Middle: Baseline fingertip leakage. Right: YCrCb skin exclusion cleans egg boundary.



Figure 7: 16-frame Video Tracking Contact Sheet tracking persistent identities through rotations and recoveries.

6. Interview Preparation & Technical Discussion

Why classical segmentation struggled on boiling pots and white trays

Global and adaptive thresholding methods fundamentally rely on contrast gradients. On white styrofoam trays and white bowls, the contrast gradient between the eggshell and surface is near-zero ($\Delta L^* \approx 0–5$). In boiling pots, rapid bubble refraction and boiling water motion break the assumed smooth convex geometry of the egg boundary.

Why Hungarian matching was chosen over DeepSORT

Eggs are smooth, rotationally symmetric, textureless objects. Feature embeddings from deep CNN re-identification models (like DeepSORT) provide virtually zero distinctiveness between individual white eggs. Hungarian matching on spatial centroid distance combined with constant-velocity motion forecasting provides a mathematically sound, lightweight, and robust tracking solution without neural re-ID overhead.

What would you do next with more time?

1. **Deep Semantic Segmentation:** Fine-tune a YOLOv8-seg or Mask2Former architecture on the dev split using SAM pseudo-masks. Deep CNNs encode contextual priors that effortlessly separate eggs from white trays.
2. **Active Contours / Snakes:** Apply Chan-Vese energy minimization with an elliptical shape prior to snap onto faint contrast boundaries on low-contrast trays.
3. **Optical Flow Association:** Integrate dense Farneback optical flow into the video tracker to track eggs through rapid hand occlusions during high-speed spinning.